MIL-B-52390A
2 December 1975
SUPERSEDING
MIL-B-52390
28 September 1964

MILITARY SPECIFICATION

BITS, AUGER; SHIP, WITH SCREW POINT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ship auger-type wood-boring bits of various sizes for use with portable power drills arranged for procurement as individual items or in a set with a canvas roll.

1.2 Classification. The auger bits shall be of the following types, sizes, classes, and style, as specified (see 6.2):

Type I - Individual auger bits.

Size A - 7/16-inch nominal diameter.
Size B - 3/4-inch nominal diameter.
Size C - 1-inch nominal diameter.
Size D - 2-inch nominal diameter.
Class 1 - 12-inch nominal length.
Class 2 - 30-inch nominal length.

Type II - Auger-bit set.

Style a - In canvas roll.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

CCC-D-950	- Dyeing and Aftertreating Processes for Cotton Cloths.
PPP-P-40	- Packaging and Packing of Hand Tools.

FSC 5133

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Military

MIL-R-45804 - Rolls, Tools and Accessories.

STANDARDS

Federal

FED. STD. No. 356 - Commercial Packaging of Supplies and Equipment.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.

DRAWINGS

ME

C13211E3416 - Bit, Auger, Size D.
C13211E3417 - Bit, Auger, Size C.
C13211E3418 - Bit, Auger, Size B.
C13211E3419 - Bit, Auger, Size A.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 Description. The auger bits shall be in accordance with the applicable drawings and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the auger bits. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3 Hole size. The diameter of holes bored with the auger bits through at least 8 inches of hardwood shall be not less than the size shown on the drawings and not more than 1/32 inch larger than the size shown on the drawings.

3.4 Bow. The auger bits shall be straight and shall be free from any bow greater than 0.015 inch when tested as specified in 4.4.3.

3.5 Identification marking. The auger bits shall be identified in accordance with MIL-STD-130. The marking shall consist of the auger bit size and the manufacturer's name or trademark applied directly to the surface by stamping or other permanent method.

3.6 Type I - Individual auger bits.

3.6.1 Size A. The Size A auger bits shall conform to Drawings C13211E3419-1 for Class 1 and C13211E3419-2 for Class 2.

3.6.2 Size B. The Size B auger bits shall conform to Drawings C13211E3418-1 for Class 1 and C13211E3418-2 for Class 2.

3.6.3 Size C. The Size C auger bits shall conform to Drawings C13211E3417-1 for Class 1 and C13211E3417-2 for Class 2.

3.6.4 Size D. The Size D auger bits shall conform to Drawings C13211E3416-1 for Class 1 and C13211E3416-2 for Class 2.

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3.7 Type II - Auger-bit set.

3.7.1 Style a. The Style a auger-bit set shall consist of a canvas roll and the following Type I auger bits:

Size A, Class 1 - 3 each.
Size A, Class 2 - 3 each.
Size B, Class 1 - 3 each.
Size B, Class 2 - 3 each.
Size C, Class 1 - 1 each.
Size C, Class 2 - 1 each.
Size D, Class 1 - 1 each.
Size D, Class 2 - 1 each.

3.7.1.1 Canvas roll. The canvas roll shall be in accordance with MIL-R-45804, Size 18, with the following exceptions:

(a) The nominal overall size of the roll shall be 49 inches long and 34 inches wide when unrolled.
(b) The roll shall be No. 10 duck, hard texture, and fitted with 16 pockets, sized to enclose the full length of the 16 auger bits of the set.
(c) The remaining length of the roll not used for pockets shall be used as a flap to cover the open end of the pockets and shall be secured to the inside of the roll with two snap fasteners.
(d) Two web straps, equipped with nonslip buckles and end clips, shall be furnished to secure the auger-bit set in the rolled position.
(e) The cloth shall be finished to conform to CCC-D-950, Type I, Class B, Inhibitor e.

3.8 Workmanship. All machined surfaces of the auger bits shall be finished as specified on the applicable drawings, and shall be free of base metal cracks, pits, or rust.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the major defects specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 4.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The auger bits shall be examined as specified herein for the following major defects:

101. Dimensions not as specified.
102. Material, heat treatment, or finish not as specified.
103. Shanks or back taper not as specified.
104. Identification marking missing, incomplete, or not legible.
105. Canvas roll construction not as specified.
106. Workmanship not as specified.

4.4.2 Tests.

4.4.2.1 Cutting. Chuck the auger bit with a chuck adapter to a 5/8-inch square drive impact hammer. Bore a hole through at least 8 inches of hardwood. Nonconformance of the hole to the tolerance specified in 3.3 or inability of the bit to bore through at least 8 inches of hardwood shall constitute failure of this test.

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4.4.2.2 Hardness. Test the auger bit for hardness in accordance with ASTM E 18 using a diamond penetrator and a 150-kilogram load or on an equivalent tester. Make measurements on opposite flats ground on the cutter head. Nonconformance to the hardness requirements specified on the drawings shall constitute failure of this test.

4.4.3 Bow. The bow shall be measured by placing the sample auger on a flat surface plate so that one inch of the auger measured from the cutter projects beyond the plate. Nonconformance to 3.4 shall constitute failure of this test.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the auger bits shall be examined to determine compliance with the quality conformance requirements of PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The auger bits shall be preserved in accordance with PPP-P-40.

5.1.2 Commercial. The auger bits shall be preserved in accordance with the requirements of FED. STD. No. 356.

5.2 Packing. Packing shall be Level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The auger bits shall be packed in accordance with the Level A requirements of PPP-P-40.

5.2.2 Level B. The auger bits shall be packed in accordance with the Level B requirements of PPP-P-40.

5.2.3 Commercial. Packing of the auger bits shall be in accordance with FED. STD. No. 356.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with FED. STD. No. 356.

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6. NOTES

6.1 Intended use. The auger bits are to be used for boring through heavy timbers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, size, class, and style, as applicable, required (see 1.2).
(c) Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Classification change. The Type II, Style b, auger-bit set in wood or metal boxes has been discontinued since it is no longer required.

Custodians:

Army - ME
Air Force - 84

Review activity:

Army - GL

User activities:

Army - EL
Navy - MC

Preparing activity:

Army - ME

Project No. 5133-0040

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-B-52390A Bits, Auger; Ship, with Screw Point

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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